

ElGamal Cryptography

Announced in 1984 by
T. Elgamal

Public-key scheme
based on discrete
logarithms closely
related to the Diffie-
Hellman technique

Used in the digital
signature standard
(DSS) and the S/MIME
e-mail standard

Global elements are a
prime number q and a
which is a primitive
root of q

Security is based on
the difficulty of
computing discrete
logarithms

Global Public Elements	
q	prime number
α	$\alpha < q$ and α a primitive root of q

Key Generation by Alice	
Select private X_A	$X_A < q - 1$
Calculate Y_A	$Y_A = \alpha^{X_A} \bmod q$
Public key	$\{q, \alpha, Y_A\}$
Private key	X_A

Encryption by Bob with Alice's Public Key	
Plaintext:	$M < q$
Select random integer k	$k < q$
Calculate K	$K = (Y_A)^k \bmod q$
Calculate C_1	$C_1 = \alpha^k \bmod q$
Calculate C_2	$C_2 = KM \bmod q$
Ciphertext:	(C_1, C_2)

Decryption by Alice with Alice's Private Key	
Ciphertext:	(C_1, C_2)
Calculate K	$K = (C_1)^{X_A} \bmod q$
Plaintext:	$M = (C_2 K^{-1}) \bmod q$

Figure 10.3 The ElGamal Cryptosystem

ElGamal Example

- Choose $q=19$ and $a=10$
- Alice computes her key:
 - A chooses $x_A=5$ & computes $y_A=10^5 \bmod 19 = 3$
- Bob send message $m=17$ as $(11,5)$ by
 - choosing random $k=6$
 - computing $K = (y_A)^k \bmod q = 3^6 \bmod 19 = 7$
 - computing $C1 = a^k \bmod q = 10^6 \bmod 19 = 11$;
 - $C2 = KM \bmod q = 7 \cdot 17 \bmod 19 = 5$
- Alice recovers original message by computing:
 - recover $K = C1^{x_A} \bmod q = 11^5 \bmod 19 = 7$
 - compute inverse $K^{-1} = 7^{-1} = 11$
 - recover $M = C2 K^{-1} \bmod q = 5 \cdot 11 \bmod 19 = 17$